Media-in-the-loop holography: designing photopolymer exposures through a differentiable recording model: supplemental document

Abstract: This document supports the main article with the full derivation behind the heuristic small-signal transfer function and the K_c prediction, the complete mechanism factorial including its unmatched conditions, the slice-count and slant tables, the miscalibration, compute-matched, target and noise studies, the RCWA check of the closed-form readout, the twin fits with the dye-bleaching rate held, implementation checks, and a map from every reported result to the files and scripts that produce it. The main article is self-contained; nothing here is needed to follow it. Section, figure and table numbers without an "S" refer to the main article.

S1. Heuristic small-signal transfer function and K_c

Derivation. The main article states the result as its Eq. (7); this is the derivation. Take $I(x) = I_{\text{mean}}(1 + m\cos Kx)$ with $m \ll 1$, neglect dye depletion ($d \approx 1$) and hold $D_{\text{eff}} = D_0$, i.e. linearize Eqs. (2)–(4) about the start of recording. The uniform fields obey $\dot{u}_0 = -F_0 u_0$ with $F_0 = \kappa I_{\text{mean}}^{\gamma}$. Eliminating the first-order Fourier component quasi-statically gives

$$H(K) = \frac{\exp(-K^2\sigma^2/2)}{1 + D_0 K^2/F_0}, \tag{S1}$$

implemented as `NPDDRecorder.small_signal_mtf` in the released code. A compensation limit follows if the pre-boost $1/H(K)$ needed to undo the roll-off exceeds the contrast budget.

Status. Equation (S1) does not match the full simulator, while an independent numerical linearization of the same equations does. $H(K)$ is therefore a heuristic that motivated the hypothesis, not a validated transfer function, and no result depends on its being exact. LPC (Section 3), which uses it directly, improves on media-blind SGD by at most 0.07 dB.

Definition of K_c. The pipeline compares $1/H(K)$ with the contrast the optimizer actually realized at each grid K, not the nominal budget, and reports the smallest K at which $1/H$ exceeds it, by linear interpolation. The mean realized contrast is 2.00, 3.80 and 5.32 at nominal budgets 2, 4 and 8, giving $K_c = 4.22$, 6.96 and 9.33 rad/µm. At budget 8 the optimizer does not realize the nominal contrast at several K, which is why K_c grows less than the nominal budget would suggest. The observed zero crossing exists only at budget 2 ($K^* = 4.18$ rad/µm).

Exposures and index profiles behind Fig. 2. Figure S1 shows the exposures $E(x)$ and recorded index profiles $\Delta n(x)$ of the reconstructions in Fig. 2. In these single-seed examples the linear assumption drives BSGD's exposure toward near-binary swings between zero and the contrast cap, which at the higher K push $\Delta n(x)$ toward $\Delta n_{\max}$ over most of the period; MIL's exposure is less extreme at the lowest K. This is an illustration of one seed, not a statistical claim.

S2. Mechanism ablations

One mechanism at a time. The conditions are $\sigma \to 0$ (no non-locality), $D_0 \to 0$ (no diffusion), $k_{\text{bleach}} \to 0$ (no dye bleaching), and the tanh index map replaced by its linear limit $\Delta n = 1.5\,\Delta n_{\max} N$ (no index saturation). Table 3 lists the means; Fig. S2 adds the intervals.

Factorial over the saturating mechanisms. M denotes monomer depletion (removed by holding $u \equiv 1$: polymer production continues at the local rate and the reservoir never empties), D dye bleaching ($k_{\text{bleach}} = 0$ when removed) and T the tanh index map (linearized when removed). "Matched" rows rescale κ so that the polymer cured at unit uniform dose is $N = 1$, as in the full

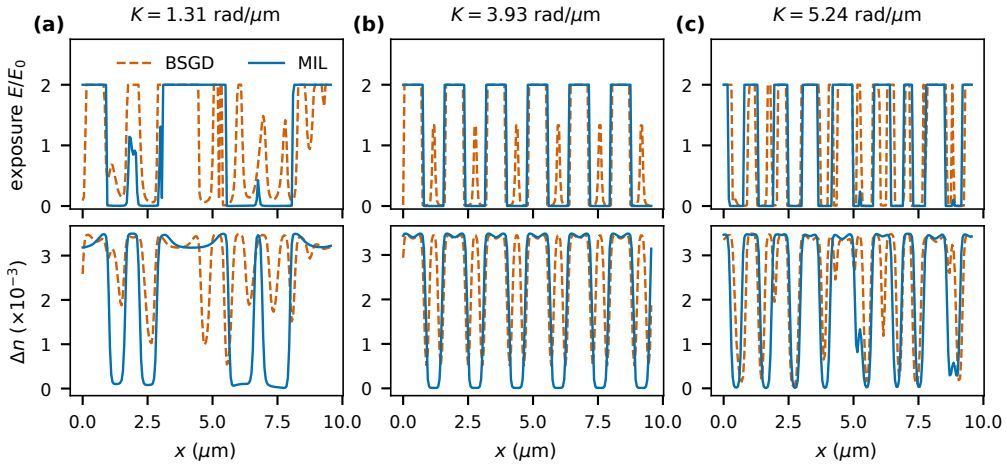


Fig. S1. Exposure E/E_0 (top) and recorded Δn (bottom) for BSGD and MIL, same K, budget $B_c = 2$, seed and 9.6 μm section as Fig. 2.

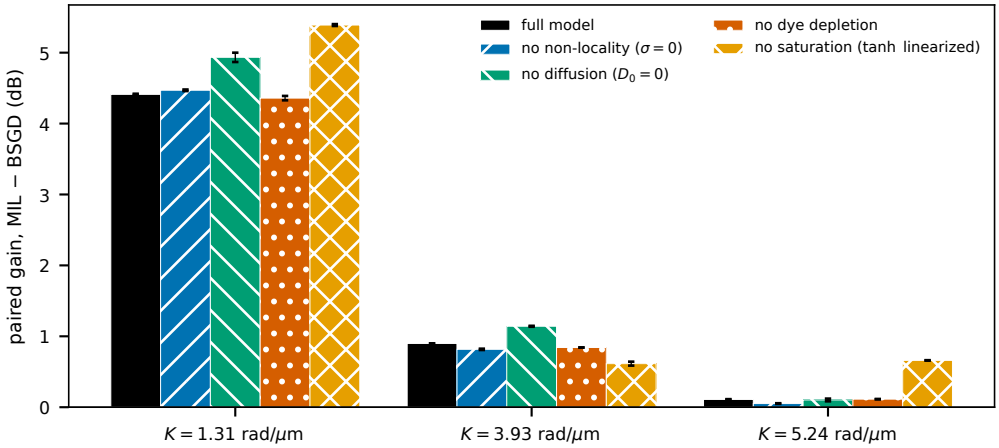


Fig. S2. Paired gain with one mechanism removed at three K, $B_c = 2$, 95% t-intervals over three seeds.

model: $\kappa = 0.2/(1 - e^{-2})$ with bleaching and $\kappa = 0.1$ without. "Linear, slope-matched" sets $\kappa = 1/15$ with all three removed, so that $\Delta n = \Delta n_{\max}(G_\sigma * E)$, the media-blind model up to blur and shrinkage. The factorial ran on CPU (single precision); its full-model row reproduces the GPU values of Table 3 to within 0.001 dB. Table S1 lists every condition and Fig. S3 shows the intervals.

Unmatched conditions. Without operating-point matching, removing monomer depletion with the tanh map kept saturates the map almost everywhere, and both methods then fail equally (gains -0.12, -0.09 and -0.08 dB; "only tanh" behaves the same way). With the map linearized as well ("only dye bleaching", "linear"), the cured polymer and hence the index modulation are an order of magnitude deeper than BSGD's linear model assumes, and the large gains (14.94, 7.66 and 9.04 dB for "linear") measure that modulation-depth mismatch rather than saturation. This is why the main text compares only operating-point-matched conditions.

Table S1. Paired gain (dB) for every factorial condition, $B_c = 2$, mean of three seeds. Active mechanisms: M monomer depletion, D dye bleaching, T tanh map.

Active	Condition	$K = 1.31$	$K = 3.93$	$K = 5.24$
M D T	full model	4.41	0.90	0.11
D T	no monomer depletion	-0.12	-0.09	-0.08
D T	matched	3.64	0.72	0.08
M T	no dye bleaching	4.35	0.84	0.11
M D	no index saturation	5.38	0.62	0.66
T	only tanh	-0.10	-0.02	-0.03
T	matched	2.80	0.53	0.12
D	only dye bleaching	14.37	6.38	7.40
D	matched	1.98	0.42	-0.02
M	only monomer depletion	5.75	1.73	1.26
–	linear	14.94	7.66	9.04
–	linear, slope-matched	-0.00	0.00	0.10

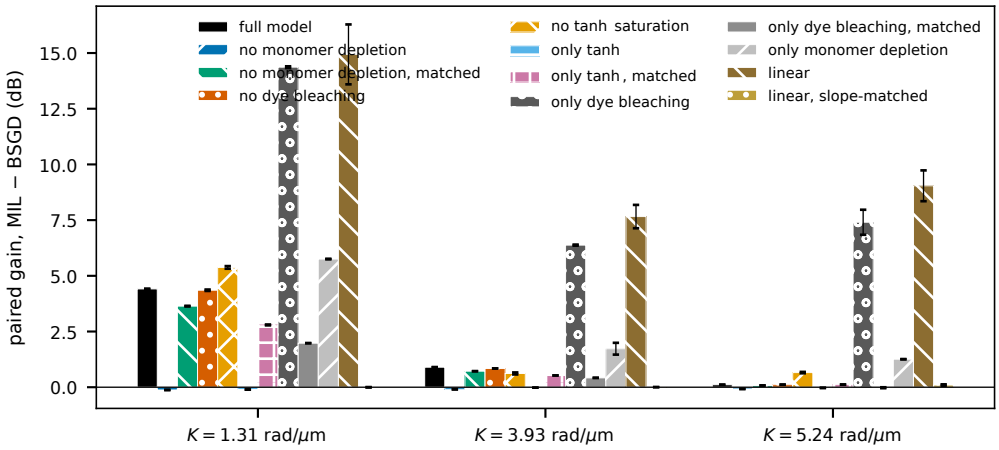


Fig. S3. Paired gain for every factorial condition of Table S1, three K, $B_c = 2$, 95% t-intervals over seeds.

S3. Readout slice count and grating slant

Table S2 gives the paired gain for $n_z = 32$, 128 and 256 (one seed, $B_c = 2$, unslanted). All experiments use $n_z = 128$; $n_z = 32$ is not converged at $K = 3.93$ rad/µm. Table S3 gives the slant dependence of Fig. 3(b) with intervals. The slanted values rely on the scalar readout outside the geometry in which it was checked against RCWA (Section S5) and are model-dependent; the intermediate-slant RCWA sweep behind Fig. 3(c), which shows the closed-form reference losing validity at the smallest nonzero tested angle, is produced by `experiments/rcwa_slant_sweep.py`.

Table S2. Paired gain (dB) versus number of readout slices n_z, one seed.

K (rad/µm)	$n_z = 32$	$n_z = 128$	$n_z = 256$
1.96	1.019	0.915	0.894
3.93	0.731	0.900	0.928
pooled over three K	0.640	0.641	0.645

Table S3. Paired gain (dB) versus grating slant ϕ, $B_c = 2$, mean of three seeds with 95% t-interval.

ϕ	0°	2.5°	5°	7.5°	10°	15°	20°
$K = 1.96$	0.913	3.122	2.077	0.227	0.089	-0.007	0.024
	[0.906, 0.920]	[3.122, 3.123]	[2.076, 2.078]	[0.223, 0.230]	[0.089, 0.089]	[-0.016, 0.002]	[0.019, 0.028]
$K = 3.93$	0.899	1.017	2.045	0.291	0.185	0.046	0.011
	[0.898, 0.901]	[0.980, 1.055]	[2.044, 2.046]	[0.210, 0.371]	[0.182, 0.187]	[0.045, 0.046]	[0.011, 0.011]

S4. One-parameter miscalibration

Both exposures are designed on the nominal twin and recorded, unchanged, on a twin with one of D_0, σ, κ or $\Delta n_{\max}$ perturbed, at $K = 1.31$, 3.93 and 5.24 rad/µm, $B_c = 2$; the gain is averaged over K for each seed and the 95% interval is taken over seeds (Fig. S4). The main article carries the compact tornado view of the same tier as its Fig. 4(a). Within ±50% the gain stays between 1.17 and 1.92 dB, the lowest value belonging to $\Delta n_{\max}$ at −50% (nominal 1.81 dB). For $\Delta n_{\max}$ the sweep is extended to bracket the 5.7× disagreement between the two Bayfol fits with k_{bleach} held (Section S9): the gain is 0.38 dB at −83%, 1.17 dB at −50%, 1.66 dB at +50%, 0.99 dB at +100% and -3.73 dB at +473%. The grid has no points between +100% and +473%, so where the sign changes is not resolved. BSGD's design depends on the medium only through the scale $\Delta n_{\max}$ of its linear model, so its exposure is the same at every perturbation; the change in paired gain reflects how MIL's design and both recordings respond to the error.

S5. RCWA check of the closed-form readout

Kogelnik first-order efficiency [1] was compared with RCWA [2], computed with TORCWA [3]. A three-case check (TE, unslanted, near Bragg incidence) agrees to 0.038 in absolute efficiency. Over the 90-case grid (TE and TM, three geometries, several Δn and K) the deviation is at most 0.29 for unslanted geometries, with median 0.001 at normal incidence, and up to 0.98 for a 20° slant, where the slant detunes the Bragg condition (Fig. S5). This checks the closed-form reference rather than the beam propagation inside the design loop; the two share the scalar approximation, which is why the headline claims are restricted to unslanted gratings.

S6. Compute-matched control

BSGD receives $\lceil 800 \times 21.7 \rceil$ iterations and MIL keeps 800. The ratio is one wall-clock measurement of per-iteration cost, not a FLOP count. The control was run at $B_c = 2$ only, three seeds, three K (Table S4); the last row gives the saturation-only surrogate's gain as a percentage of MIL's at the same points.

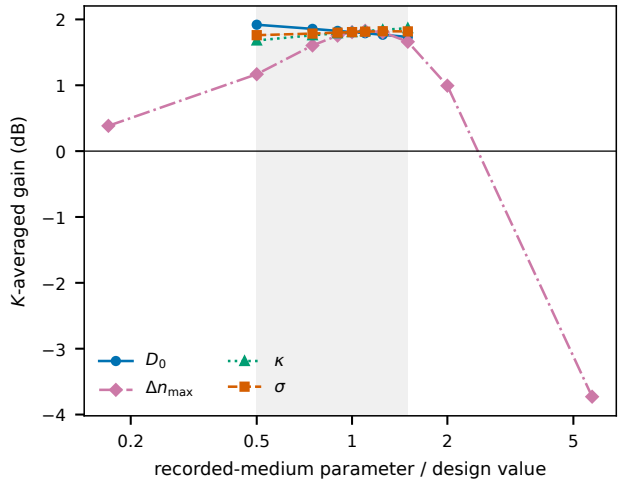


Fig. S4. Paired gain when the exposure is designed on the nominal twin and recorded on a twin with one parameter scaled, K-averaged over the three ablation K, $B_c = 2$; shading, 95% intervals over seeds; grey band, ±50%. The main article's Fig. 4(a) summarizes the same data as a tornado ranking.

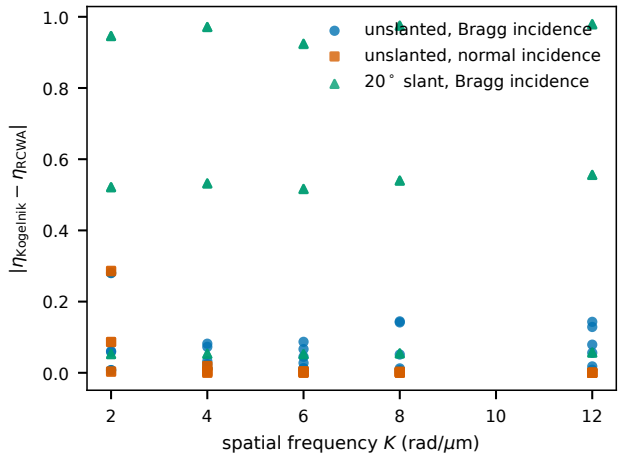


Fig. S5. Absolute deviation of Kogelnik first-order efficiency from RCWA over the 90-case grid, one series per geometry.

Table S4. Compute-matched paired gain (dB), $B_c = 2$.

	$K = 1.31$	$K = 3.93$	$K = 5.24$
MIL − BSGD (compute-matched)	3.19	0.81	0.12
95% interval	[2.94, 3.43]	[0.81, 0.81]	[0.12, 0.13]
MIL − BSGD (equal iterations)	4.41	0.90	0.11
SAT as % of MIL	79	56	-28

S7. Calibrating the twin for a new medium

Section 6 of the main article states the protocol; these are the fit-level observations it rests on. From the fits of Section 5 and Section S9: (i) κ and $\Delta n_{\max}$ are identifiable from one Δn_1(dose) curve; (ii) D_0 and σ act through the spatial-frequency dependence and need a multi-K family or an independent measurement, so a single-K curve should not be used to fit them; (iii) $\Delta n_{\max}$ must come from the same experimental condition as the curve being fitted, not from a separate experiment on the same material; (iv) the dose-to-time mapping and k_{bleach} are coupled, so k_{bleach} should be fitted (as in Section 5) or the source intensity recovered; holding it at an assumed value can make the fit degenerate (Section S9).

S8. Target content and readout noise

At $K = 3.93\,\text{rad/µm}$, three target families (bars, sparse random spots, random binary), budgets 2 and 8, three seeds, with bootstrap 95% intervals [4] (Table S5). Gaussian detector noise at 5% of the reconstructed peak, applied identically to every method before scoring, changes the bar-target gain from 0.90 to 0.88 dB ($B_c = 2$). With three seeds the bootstrap intervals are coarse; they are reported alongside, not instead of, the t-intervals used elsewhere.

Table S5. Paired gain (dB) by target family at $K = 3.93\,\text{rad/µm}$, with bootstrap 95% intervals.

Target	$B_c = 2$	$B_c = 8$
bars	0.90 [0.90, 0.90]	1.95 [1.95, 1.95]
sparse spots	3.22 [3.19, 3.23]	4.07 [3.99, 4.12]
random binary	0.17 [0.17, 0.17]	0.24 [0.24, 0.24]

S9. Twin fits with the dye-bleaching rate held

Figure S6 shows the fits with only κ and $\Delta n_{\max}$ free and $k_{\text{bleach}} = 0.2$ held (best of ten random starts). For the Bayfol HX model curve the fit reproduces the rise and decline (NRMSE 0.17). For the measured series the optimum (NRMSE 0.28) is degenerate: the fitted κ saturates the medium before the first measured dose, so the twin returns an almost constant Δn_1, and the fitted $\Delta n_{\max}$ (0.289) is 5.7× that of the model-curve fit (0.0504). Freeing k_{bleach} removes the degeneracy (Fig. 5). The PQ/PMMA curve of Hsieh et al. [5] is the source's own kinetic-model output at $K = 24.94\,\text{rad/µm}$, outside the tested range, and is still rising at the end of the record, so $\Delta n_{\max}$ is under-constrained; its fit (NRMSE 0.10) is not used as evidence in the main text.

S10. Implementation checks

Non-local kernel. The kernel G_σ is applied to the production term Fu (polymer initiated at x' is deposited at x), not to u alone, as the NPDD model requires [7].

Conservation. The explicit reaction step caps consumed monomer at the monomer available, so $u \geq 0$ and $u + N$ is conserved to rounding error at any step size.

Gradients. At the configuration of the main grid ($n_x = 1024$, 300 integrator steps, $n_z = 128$, default medium, $B_c = 2$, bar target at $K = 3.93\,\text{rad/µm}$, float64), the directional derivative of the loss from reverse-mode differentiation through the full chain (softplus, projection, recording, beam propagation, loss) agrees with a central finite difference along 6 random unit directions

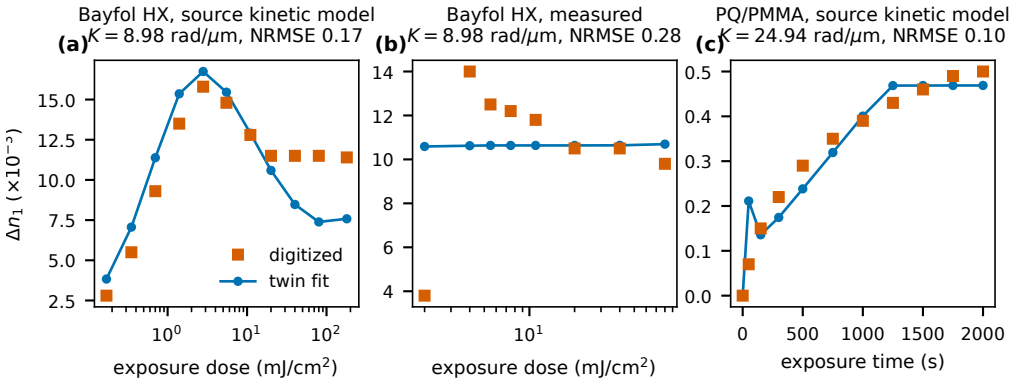


Fig. S6. Twin fits with $k_{\text{bleach}} = 0.2$ held: (a) Bayfol HX model curve and (b) measurement [6]; (c) PQ/PMMA model curve [5], outside the tested K range.

to a maximum relative error of 6.5×10^{-6} (`experiments/gradient_check.py`; the test exposure is kept below the contrast cap so the projection is smooth there).

Reproducibility. Each per-job result file records its configuration, the git commit of the code that produced it and the device; the seed enters only through the optimizer initialization.

S11. Provenance of reported results

Every number in the main article and this supplement is a macro in `numbers.tex`, generated by `scripts/make_numbers_tex.py` (with `analysis/de_confirmation.py` and `analysis/wasted_media.py`) from the per-job result files in `results/`, one JSON file per tier, configuration, method and seed, aggregated by `analysis/aggregate.py`. Figures are generated by `figures/make_all.py`. Table S6 maps each result to its source; paths are relative to `part2/code/` of the released repository, and `REPRODUCE.md` there lists the commands.

References

1. H. Kogelnik, "Coupled wave theory for thick hologram gratings," Bell Syst. Tech. J. **48**, 2909–2947 (1969).
2. M. G. Moharam and T. K. Gaylord, "Rigorous coupled-wave analysis of planar-grating diffraction," J. Opt. Soc. Am. **71**, 811–818 (1981).
3. C. Kim and B. Lee, "TORCWA: GPU-accelerated Fourier modal method and gradient-based optimization for metasurface design," Comput. Phys. Commun. **282**, 108552 (2023).
4. B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap* (Chapman & Hall, 1993).
5. Y.-H. Hsieh, Y.-C. Cheng, and T.-Y. Chung, "Study on PQ/PMMA photopolymer films fabricated by solvent casting," ACS Omega **7**, 11770–11776 (2022).
6. F.-K. Bruder, T. Fäcke, and T. Rölle, "The chemistry and physics of Bayfol HX film holographic photopolymer," Polymers **9**, 472 (2017).
7. J. T. Sheridan and J. R. Lawrence, "Nonlocal-response diffusion model of holographic recording in photopolymer," J. Opt. Soc. Am. A **17**, 1108–1114 (2000).

Table S6. Source of each reported result.

Result	Tier and source
Paired gain on the grid, Figs. 1(a) and 3(a), failure rates (Table 2), diffraction efficiency	M1; `aggregate.py`, `wasted_media.py`, `de_confirmation.py`
Baselines and SAT, Fig. 1(b)	M1
Reconstructions and profiles, Figs. 2 and S1	`experiments/make_r1_reconstructions.py`, `make_r1_profiles.py`
Compute-matched control	M2R
Ablations and factorial, Table 3, Tables S1, Figs. S2–S3	S1, S1X
Slant and slice count, Fig. 3(b), Tables S2–S3	S8, NZ0
Intermediate-slant RCWA sweep, Fig. 3(c)	`experiments/rcwa_slant_sweep.py` → `results/applied_optics/rcwa_slant_sweep.json`
Miscalibration, joint draws, depth absorption, Fig. S6	S3, S6, S7 (`experiments/run_s3_mismatch.py`, `run_s6_joint_mismatch.py`, `run_s7_depth_absorption.py`)
Parameter sensitivity ranking, Fig. 4	S3, S6 via `analysis/sensitivity.py` → `results/applied_optics/sensitivity.json`
Re-optimized parameter sensitivity	S2R
Targets and noise, Table S5	S4, S5
Twin fits and held-out prediction, Figs. 5 and S5	`experiments/fit_literature_curves.py`, `fit_twin_holdout.py`; digitized data in `data/literature/`
RCWA check, Fig. S4	`experiments/rcwa_crosscheck.py`
Medium and simulation parameters (Table 1)	`experiments/manifest.py`, `configs/media/`